



Josh Bleskacek

PORTFOLIO

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Hi, I'm Josh

My passion for engineering stems from my love of turning my ideas into physical solutions. My dream is to develop complex hardware systems that drive technological progress.

Wisconsin Racing

FORMULA SAE

Fall 2025 – Present

Overview:

Wisconsin Racing competes in the EV class of the Formula SAE competition. I am a member of the chassis subteam which is responsible for suspension and controls (steering, brakes, etc.).



Suspension Bushings

- Suspension Bushings – Spring 2026
Fabricated more than 150 spacer bushings for retaining spherical bearings on the suspension rods. Optimized and edited outdated CNC lathe programs in Mastercam.
- Various Fabrication – Spring 2026
CNC machined the clevises for the rear wing struts. Aided aero team with front wing carbon fiber layups. Machined suspension compression limiters.
- Steering Motor – Present
Early stages of designing the steering motor assembly for WR 227's self driving capability.



Projects:

- Racing Sim Upgrades – Fall 2025
Designed and manufactured reinforcing brackets for aluminum extrusion frame.
- Tire Stiffness Testing Jig – Fall 2025
Explored structural geometry of standard steel tubing. Simulated loads and analyzed cost.



Rear Wing Strut Clevis

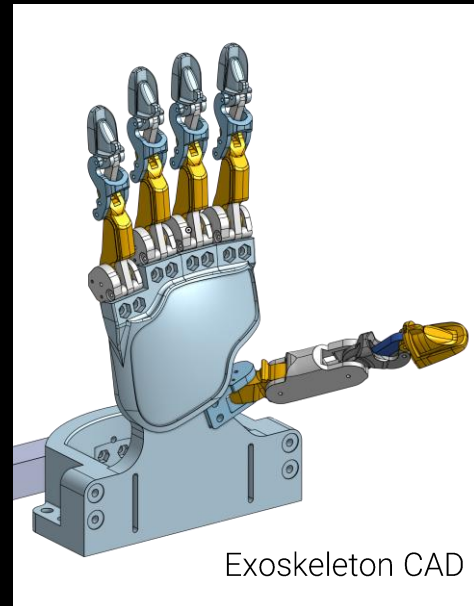
Wisconsin Humanoid Robotics



Winter 2025 – Present

Overview:

Founded in late 2025, Wisconsin Humanoid Robotics is the second humanoid robotics club in the nation, endeavoring to explore and advance this emerging field. I am on the hands subteam which is responsible for the humanoid's hands, wrists, and the teleoperated controls for these systems.



Exoskeleton CAD



Projects:

- **Exoskeleton – Spring 2026**
Led a small team to develop an exoskeleton for teleoperated control and policy training of an open-source robotic hand. Replicated robotic hand contact patches and kinematics. Designed double linkage system in fingers to mirror the underactuated motion of hand.
- **Complex Hand Design – Present**
In the initial stages of designing a robotic hand for our next humanoid. Developing design specifications based on objective tasks.

Ripsaw Robotics

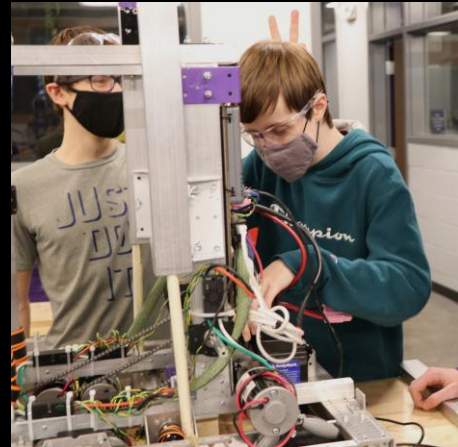
FIRST ROBOTICS COMPETITION

Design and Fabrication Lead / Team Captain
Fall 2021 – Spring 2025



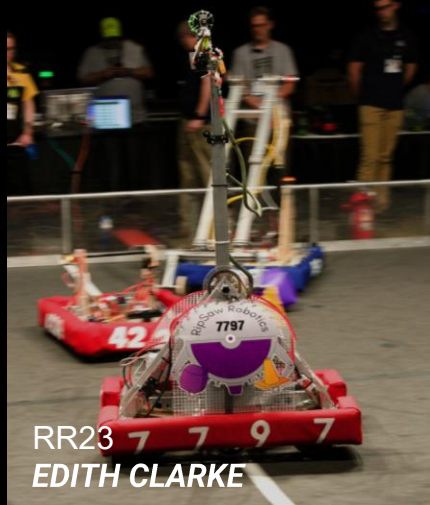
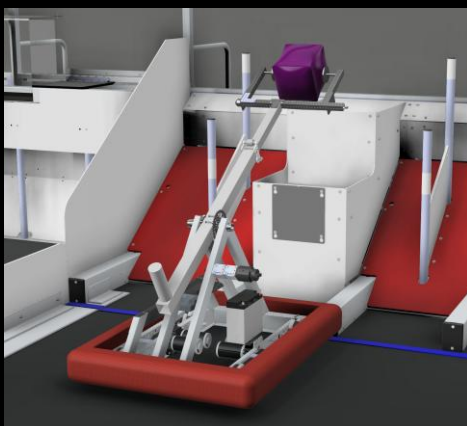
Overview:

Ripsaw Robotics is my high school's FIRST Robotics team. Over the years, I developed my and the team's skills in design, fabrication, and technical communication. In four years, we progressed from designs based on rough 3D shapes to detailed assemblies, and from eyeballed fabrication to using CNC machining, 3D printing, and welding.



Robot Evolution:

- 2022 – RR22 *Jim-Bob-Billy-Bob-Pete-Sam-Joe*
Focused on learning about FIRST robot design and the competition. Built the prototype for the ball shooter and helped with miscellaneous projects.
- 2023 – RR23 *Edith Clarke*
Led the design using Fusion 360. Learned numerous hard lessons through design flaws in the arm, intake, and a scraped drive train. These experience taught me to keep my design's simple and make design decisions based on game strategy. Fabricated the claw and parts for the arm.



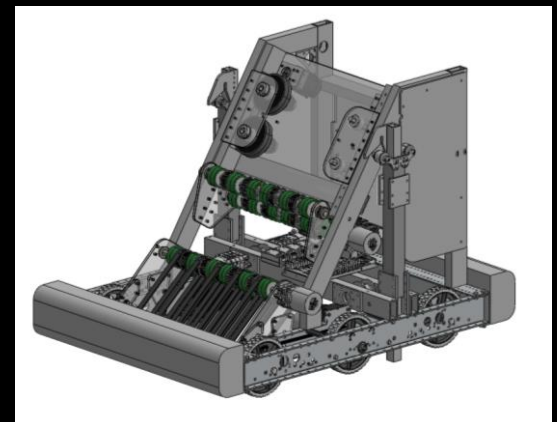
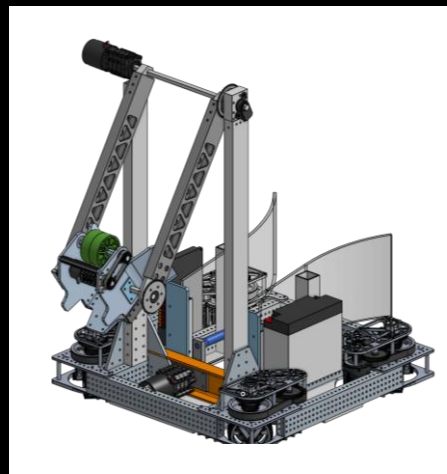
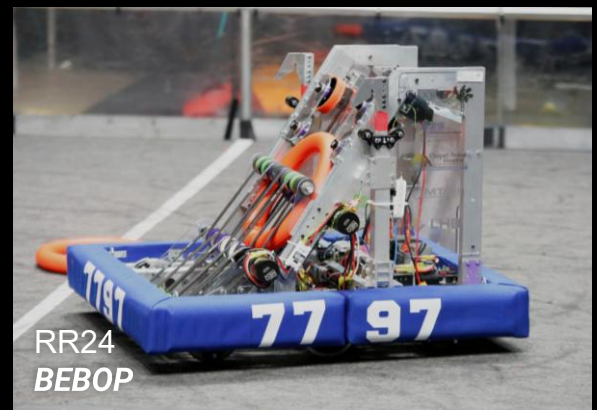
Ripsaw Robotics Continued

FIRST ROBOTICS COMPETITION



Design and Fabrication Lead / Team Captain
Fall 2021 – Spring 2025

- Summer 2023 – PepBot
The goal for PepBot was to create a fun robot for recruiting and community events. Built it over the summer in a garage with a small team. Designed its notable servo actuated googly eyes and a pencil dispenser.
- 2024 – RR24 *Bebop*
Led the design (SolidWorks) and fabrication of Bebop. The design philosophy was built around simple scoring cycles. Self taught myself CNC milling to make custom brackets for our climber. Simplified fabrication with a high utilization of milled polycarbonate. Directed strategy at competition as the drive team coach. We had the best regional performance in the team's history and made it to state for the first time.
- 2025 – RR25 *Riptide*
In the off season, I worked with a small group to implement the team's first swerve drive. Riptide was designed in Onshape its virtual four-bar linkage and an interior climbing system were unique design features. It also featured a large percentage of parts I CNC milled and plasma cut.



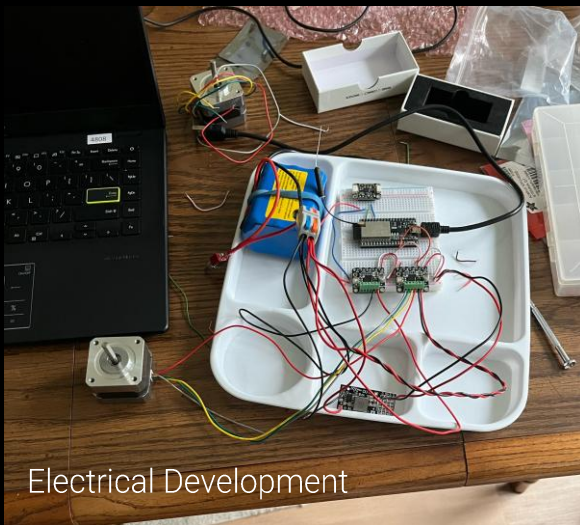
BalanceBot

SENIOR PHYSICS CAPSTONE

Spring 2025

Overview:

BalanceBot is half personal project and half physics capstone. The hardware was designed using SolidWorks and programmed with C++. It utilizes a 3D printed body and wheels. The wheels are driven by two stepper motors and controlled by A4988 motor drivers connected to an ESP32 microcontroller. To balance, the robot uses a PID controller with an angle measurement input, from a MPU6050 inertial measurement unit, and a motor output.



Crane Project

ME 201 Final

Spring 2025

Overview:

For the ME 201 Introduction to Mechanical Engineering final project groups were tasked with designing and building the lightest crane that could lift a 1.75kg weight and move it over several obstacles. My design utilized an extendable counterweight boom to avoid adding extra counterweight mass. We had a pulley fail during testing the night before

